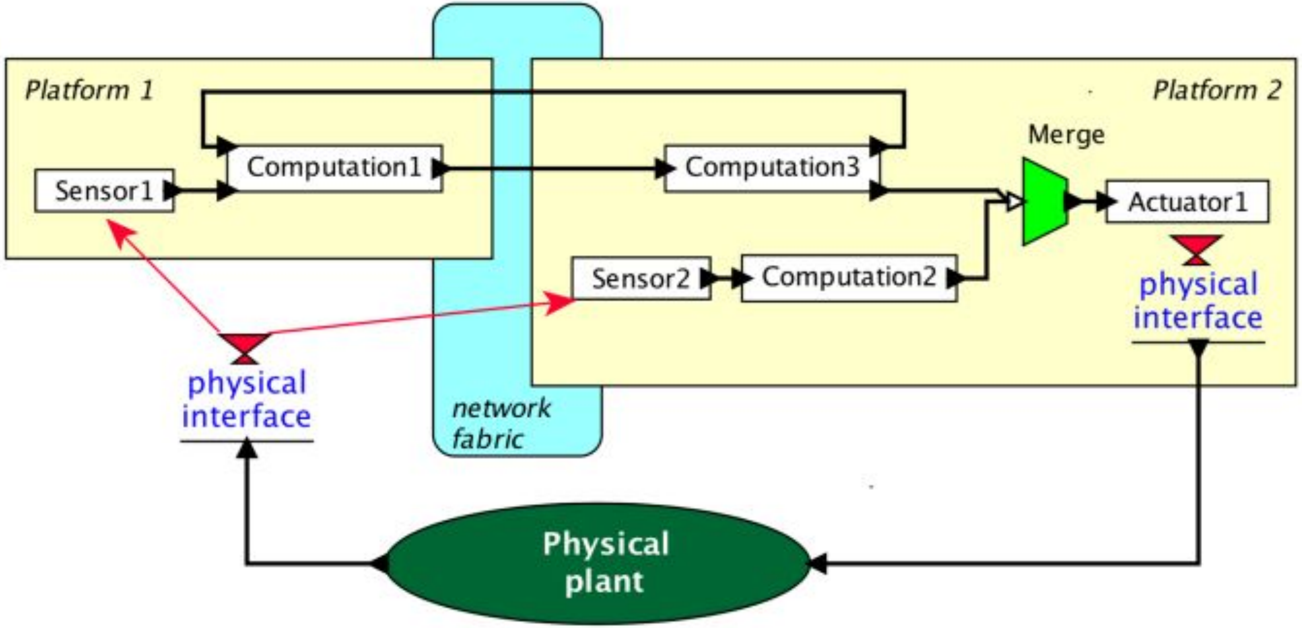
CYBER PHYSICAL SYSTEMS

Understand the Features and Components of Cyber Physical Systems

About the Term

- The term “cyber-physical systems” emerged in 2006, coined by Helen Gill at the National Science Foundation in the US.



What are Cyber-Physical Systems?

- *"Cyber-Physical Systems or 'smart' systems are co-engineered interacting networks of physical and computational components. These systems will provide the foundation of our critical infrastructure, form the basis of emerging and future smart services, and improve our quality of life in many areas."*

-- NIST, Engineering Laboratory

What are Cyber-Physical Systems? (Contd.)

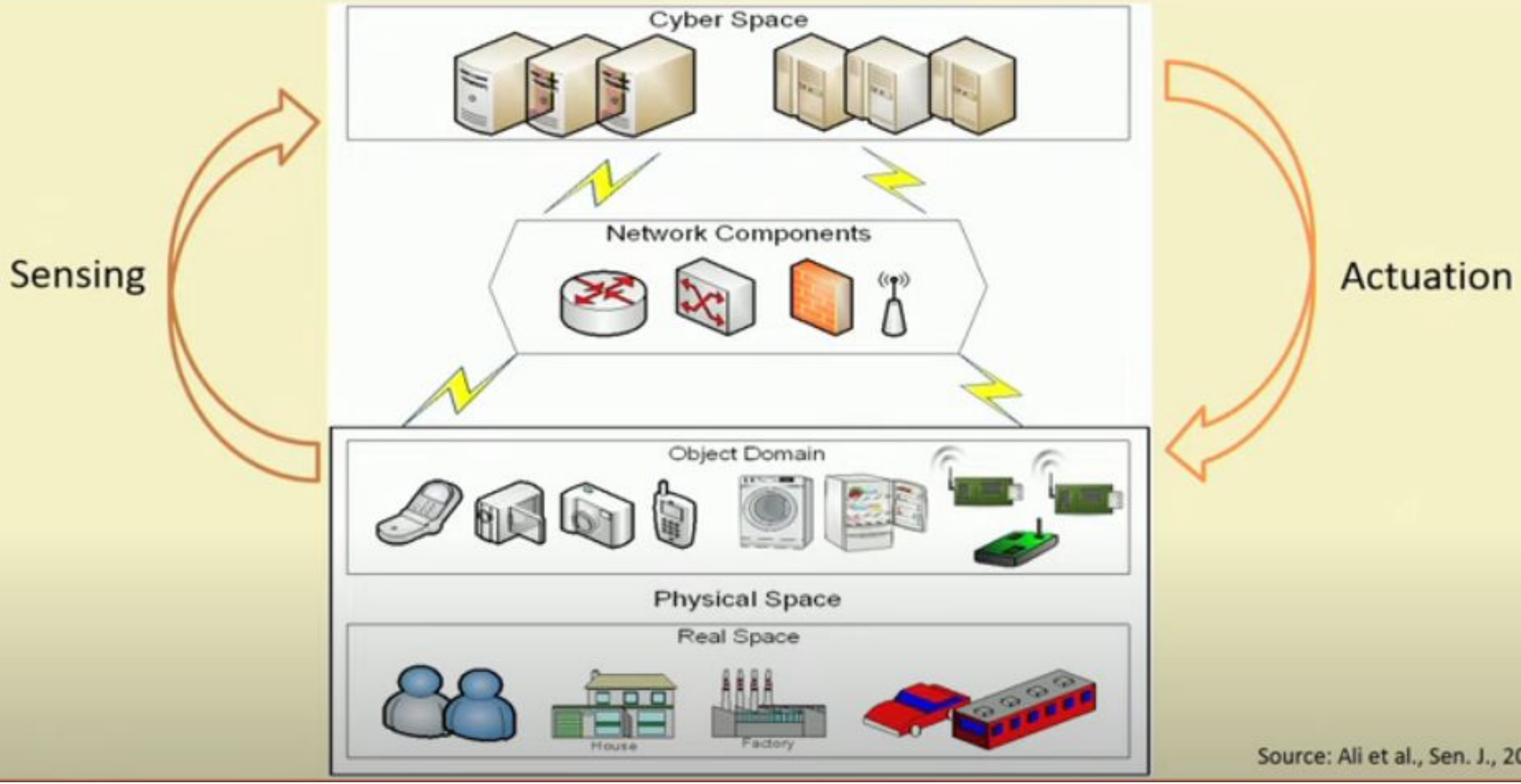
- Generalization of “embedded” systems
  - Possess *compute, communicate* and *control* capabilities
  - Interaction with the physical world through sensors and actuators.
- Examples:
  - Medical instruments
  - Transportation vehicles
  - Defense systems
  - Robotic equipment
  - Process monitoring and factory automation systems

CPS INTRO

- An embedded system consists of hardware and software integrated within a mechanical or an electrical system designed for a specific purpose.
- From watches to cameras to refrigerators, almost every engineered product today is an embedded system with an integrated microcontroller and software. The concept of a cyber-physical system is a generalization of embedded systems.
- A cyber-physical system consists of a collection of computing devices communicating with one another and interacting with the physical world via sensors and actuators in a feedback loop.
- Increasingly, such systems are everywhere, from smart buildings to medical devices to automobiles.

EXAMPLE

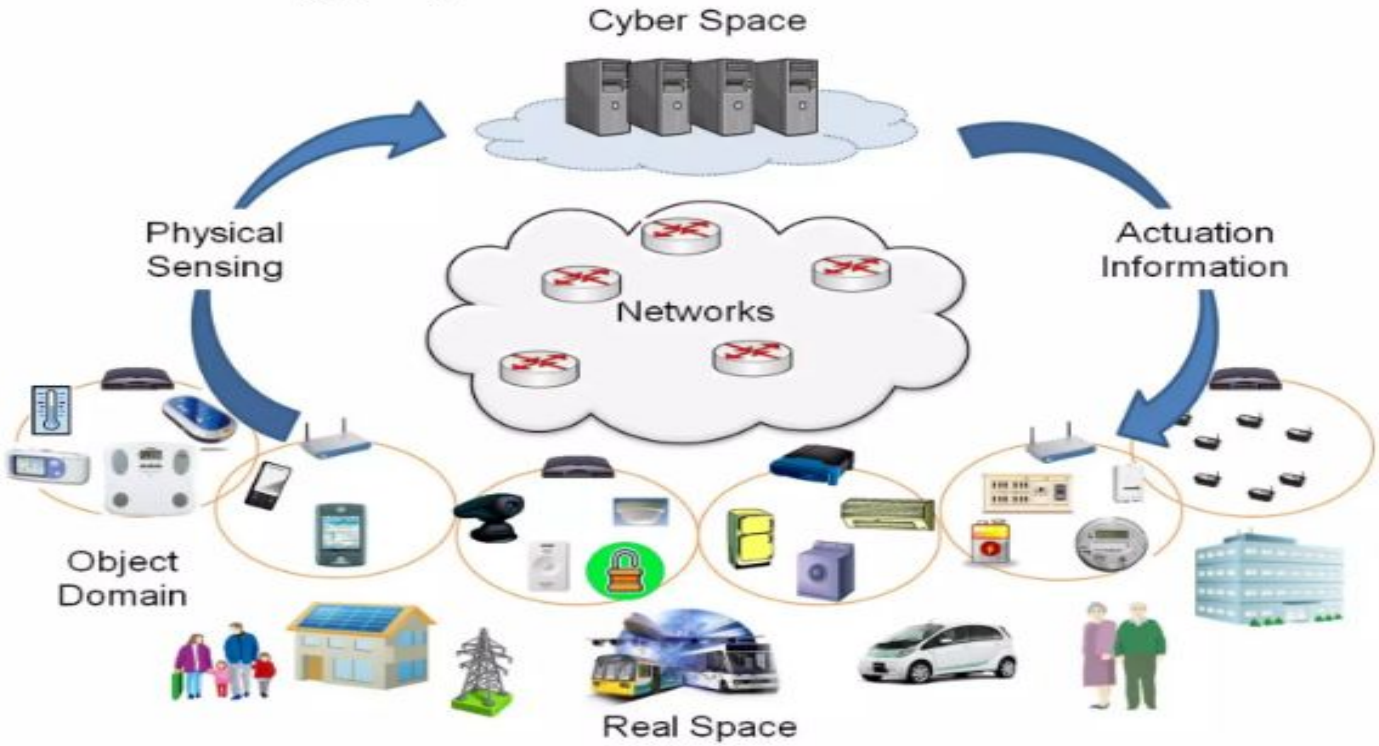
- ❖ Consider a team of autonomous mobile robots tasked with the identification and retrieval of a target inside a house with an unknown floor plan.
- ❖ To achieve this task, each robot must be equipped with multiple sensors that collect the relevant information about the physical world.
- ❖ Examples of on-board sensors include a GPS receiver to track a robot's location, a camera to take snapshots of its surrounds, and an infrared thermal sensor to detect the presence of humans.
- ❖ A key computational problem then is to construct a global map of the house based on all the data collected, and this requires the robots to exchange information using wireless links in a coordinated fashion.
- ❖ The design objectives include safe operation (for instance, a robot should not bump into obstacles or other robots), mission completion (for example, the target should be retrieved), and physical stability (for example, each robot should be stable as a dynamical system).
- ❖ Construction of the multi-robot system to meet these objectives requires design of strategies for control, computing, and communication in a synergistic manner.

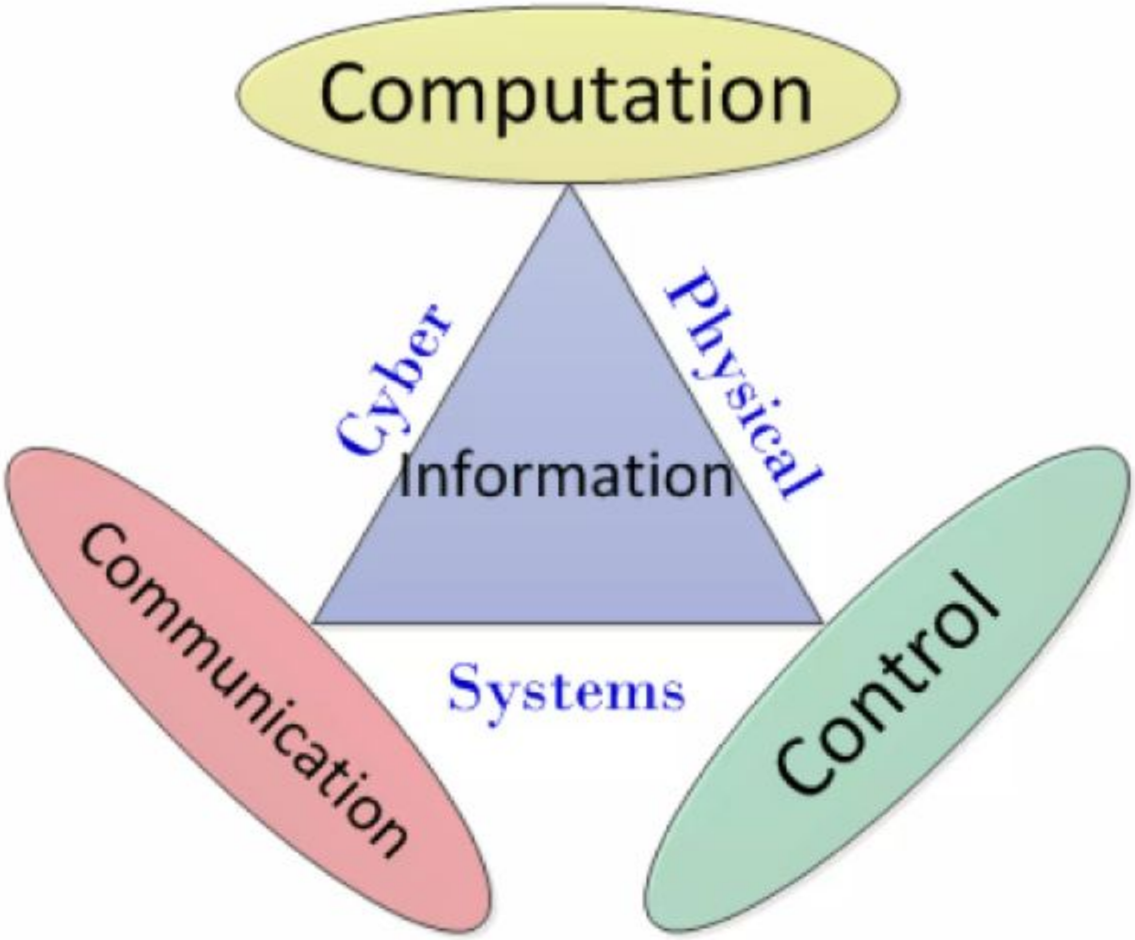


Source: Ali et al., Sen. J., 2015

Cyber-Physical Systems

- System of collaborating computational elements controlling physical entities




Computation
Cyber
Physical
Information
Communication
Control
Systems

Differences with Embedded Systems

Embedded Systems	CPS
Devices having information processing systems embedded into them	Complete system having physical components and software
Typically confined to a single device	Networked set of embedded systems
Limited resources for performing limited number of tasks	Not resource constrained
Main issues are real-time response and reliability	Main issues are timing and concurrency

Features of Cyber-Physical Systems

- Reactive Computation:
  - Interact with environment in an ongoing manner
  - Sequence of observed inputs and outputs
- Concurrency:
  - Multiple processes running concurrently
  - Processes exchange information to achieve desired result
  - Synchronous or asynchronous modes of operation

REACTIVE COMPUTATION EXAMPLE

- Consider a program for a cruise controller in a car.
- Such a program receives high-level input commands for turning the cruise controller on and off and for changing the desired cruising speed. The control program needs to respond to such inputs by changing its output, which corresponds to the force that is applied to the engine throttle.
- The behavior of such a system then is naturally described by a sequence of observed inputs and outputs, and the notion of correctness specifies which input/output sequences correspond to acceptable behaviors.

CONCURRENCY

- Synchronous models: components execute in lock-step, and the computation progresses in a logical sequence of synchronized rounds
- Asynchronous models, components execute at independent speeds, exchanging information by sending and receiving messages.

CONCURRENCY EXAMPLE

- In our example of a team of autonomous mobile robots, the robots themselves are separate entities and are thus executing concurrently.
- Each robot has multiple sensors and processors, and computing tasks such as constructing a map of the environment based on vision data and motion planning based on the map of the environment can be executing on separate processors in parallel.
- The motion planning task can be subdivided into logically concurrent subtasks such as local planning to avoid obstacles and global planning for optimal progress toward the target.

Features of Cyber-Physical Systems (Contd.)

- Feedback Control of the Physical World:
  - Equipped with *control systems* with feedback loop
  - Sensors sense environment and Actuators influence it
  - *Hybrid* control systems for complex tasks
- Real-Time Computation:
  - Time sensitive operations such as coordination, resource-allocation
- Safety-Critical Applications:
  - Precise modelling and validation prior to development

- **Example for Feedback loop**:A cruise controller is constantly monitoring the speed of the car and adjusts the throttle force so that the speed stays close to the desired cruising speed
- Design of controllers for the physical world requires modeling the dynamics of the physical quantities: to adjust the throttle force, a cruise controller needs a model of how the speed of the car changes with time as a function of the throttle force.
- Hybrid means a mix of continuous and discrete dynamics
- **Example for Real Time Computation:**For a cruise controller to satisfactorily control the speed of the car, its design should take into account the time it takes its subcomponents to execute the necessary computations and communicate the results.
- Applications where the safety of the system has a higher priority over other design objectives such as performance and development cost are called **safety-critical**.
- The more principled approach to system development involves writing mathematically precise requirements of the desired system, designing models of system components along with the environment in which the system is supposed to operate, and using analysis tools to check that the system model meets the requirements.

Applications of CPS: Healthcare

- Highly accurate medical devices and systems
  - Image-guided surgery and therapy
  - Control of fluid flow for medicinal purposes and biological analysis
  - Intelligent operating theatres and hospitals
- Engineered systems based on cognition and neuroscience (e.g., brain-machine interfaces, therapeutic and entertainment robotics, orthotics and exoskeletons, and prosthetics)

Applications of CPS: Transportation

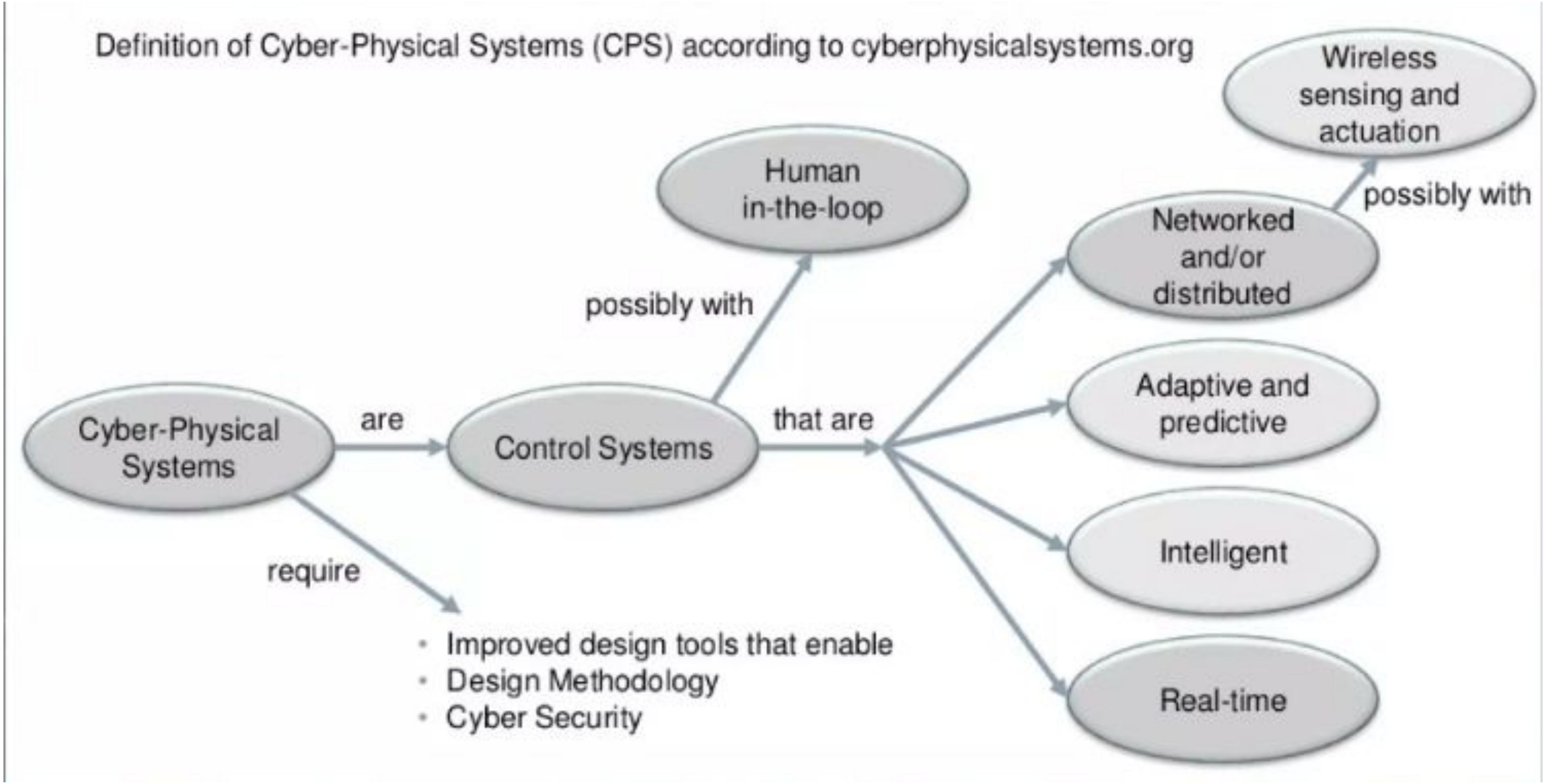
- Infrastructure-based transportation CPS
  - Real-time monitoring of traffic infrastructure (traffic signals, cameras, etc.) and traffic control
- Vehicle-Infrastructure-coordinated transportation CPS
  - Transit signal priority, queue warning (for e.g., ambulances)
- Vehicle-based transportation CPS
  - Proximity detection for safety
  - Vehicle health monitoring

Applications of CPS: Smart Grid

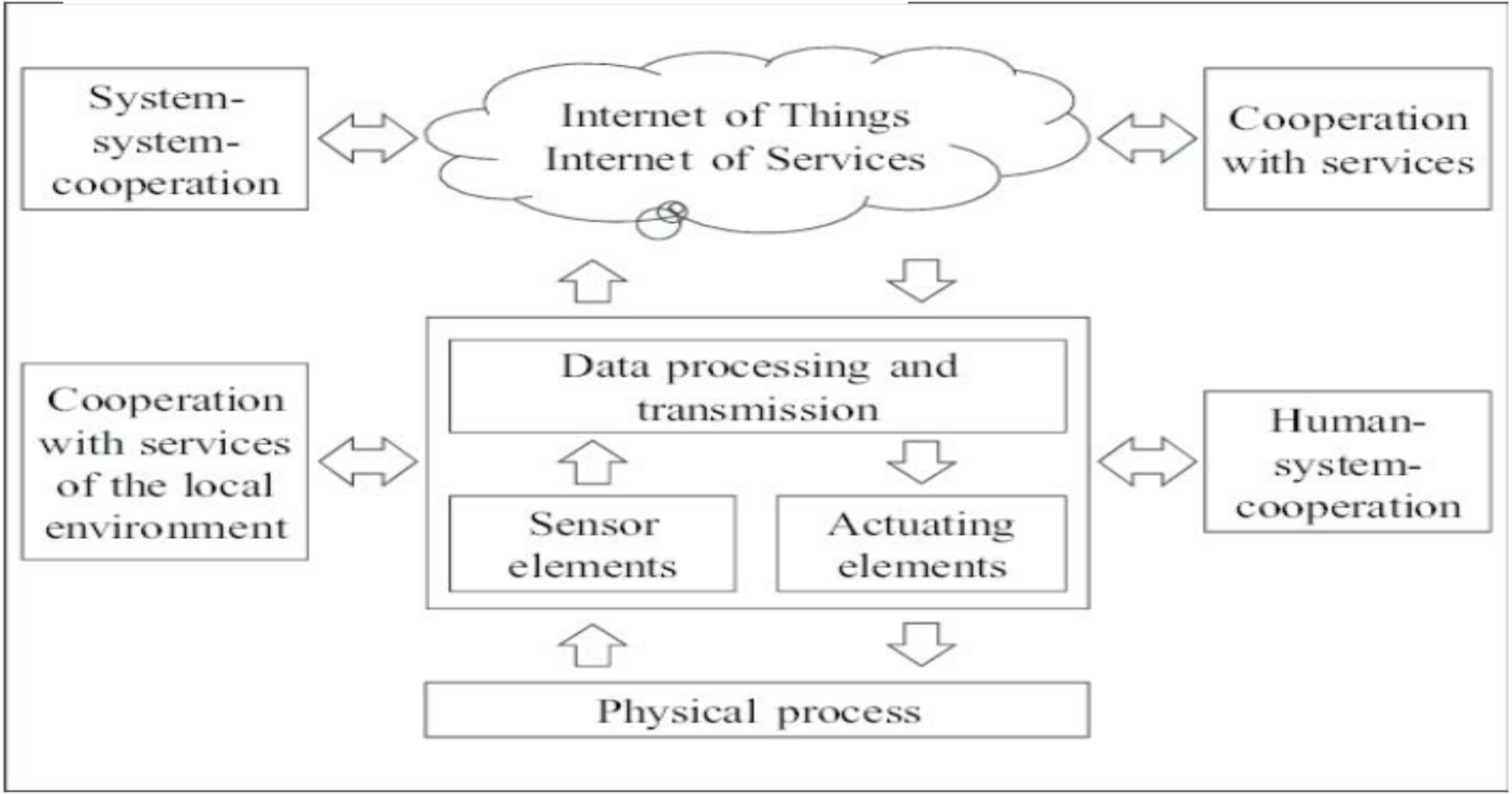
- Smart meters
  - Demand management with distributed generation
  - Automated distribution with intelligent substations
  - Wide-area control of Smart grids
- Phasor measurement units (PMUs)
- Data aggregation units (DAUs)

Applications of CPS: Industry

- Manufacturing systems and logistics integrated with communication abilities, sensors and actuators
  - Smart control
  - Optimal resource utilization
  - Smart diagnostics and maintenance
- Flexibility of development of systems
- End products customized specific to needs of customers


Definition of Cyber-Physical Systems (CPS) according to cyberphysicalsystems.org
Cyber-Physical Systems
are
Control Systems
possibly with
Human in-the-loop
that are
Networked and/or distributed
possibly with
Wireless sensing and actuation
Adaptive and predictive
Intelligent
Real-time
require
Improved design tools that enable
Design Methodology
Cyber Security

GENERAL STRUCTURE OF A CPS



Internet of Things (IoT)

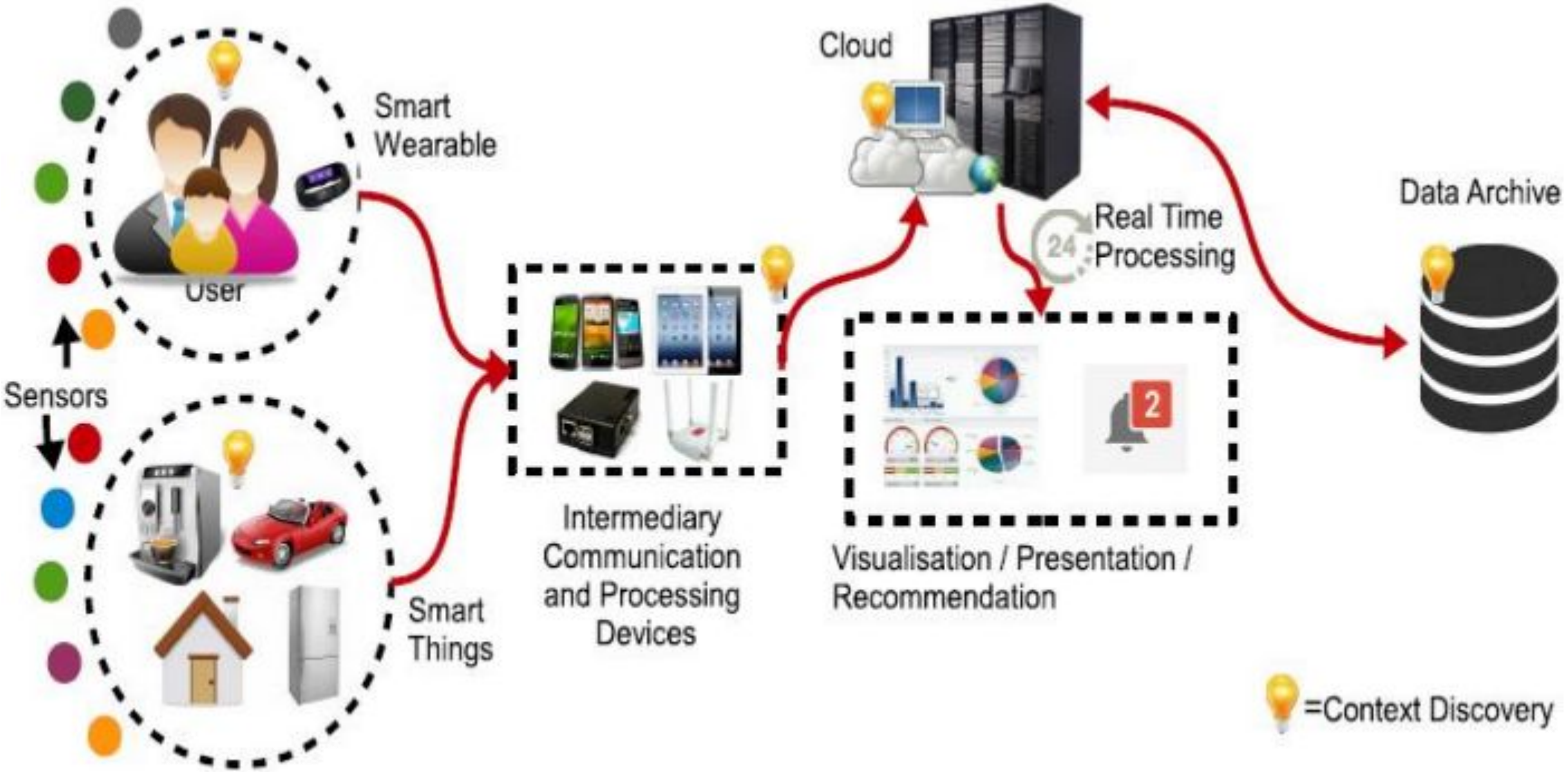
What is IOT?

- The Internet of Things is the network of physical objects or "things" embedded with electronics, software, sensors, and network connectivity, which enables these objects to collect and exchange data.

- It allows objects to be sensed and controlled remotely across existing network infrastructure, creating opportunities for more direct integration between the physical world and computer-based systems, and resulting in improved efficiency, accuracy and economic benefit.

- "Things," in the IoT sense, can refer to a wide variety of devices such as heart monitoring implants, biochip transponders on farm animals, automobiles with built-in sensors, DNA analysis devices for environmental/food/pathogen monitoring or field operation devices that assist firefighters in search and rescue operations.

IoT Ecosystem



Sensors in even the holy cow!

In the world of IoT, even the cows will be connected and monitored. Sensors are implanted in the ears of cattle. This allows farmers to monitor cows' health and track their movements, ensuring a healthier, more plentiful supply of milk and meat for people to consume. On average, each cow generates about 200 MB of information per year.

IOT Application Scenario - Shopping

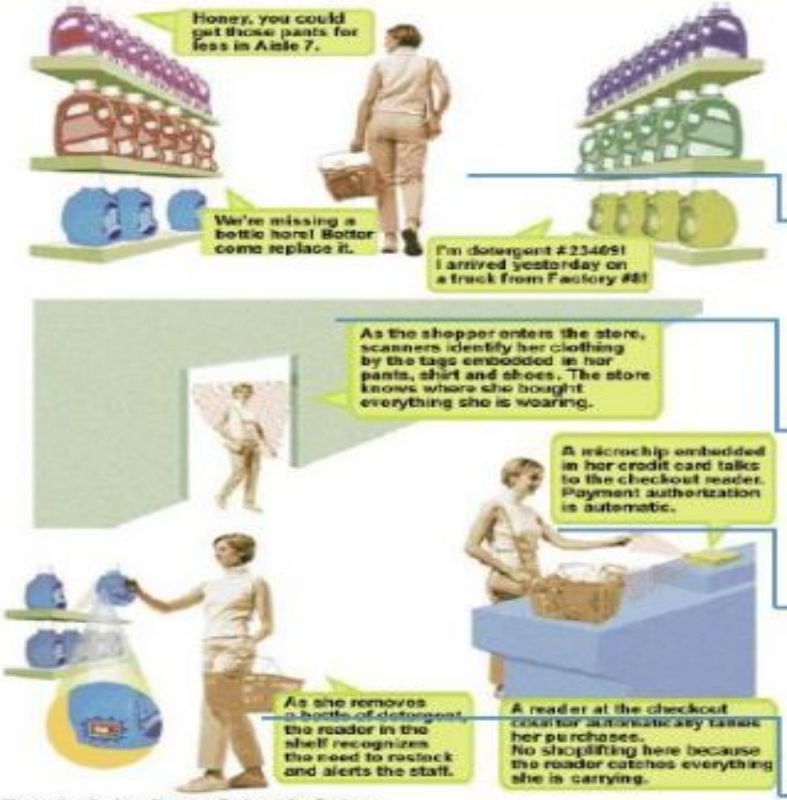


Illustration by Lisa Knouse Braiman for Forbes

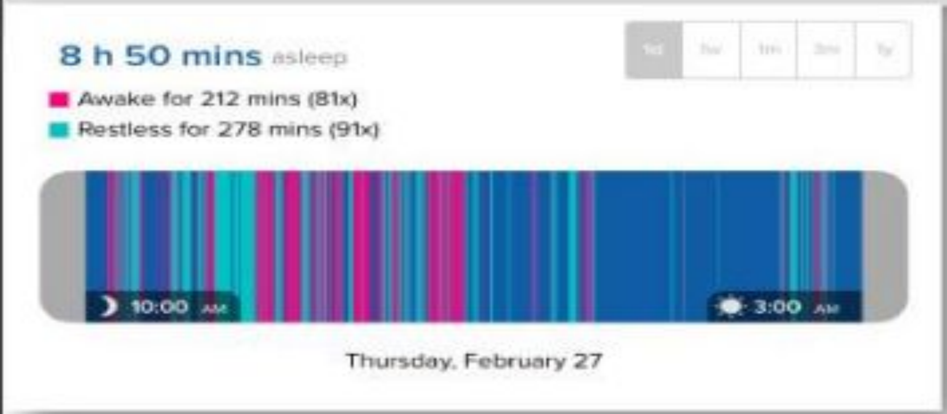
(2) When shopping in the market, the goods will introduce themselves.

(1) When entering the doors, scanners will identify the tags on her clothing.

(4) When paying for the goods, the microchip of the credit card will communicate with checkout reader.

(3) When moving the goods, the reader will tell the staff to put a new one.

How Well Do I Sleep?



fitbit flex
Wireless Activity + Sleep Wristband

CHARACTERISTICS OF IoT

1. Connectivity

Things of IoT should be connected to the IoT infrastructure. Anyone, anywhere, anytime can connect, this should be guaranteed at all times.

2. Intelligence and Identity

The extraction of knowledge from the generated data is very important. For example, a sensor generates data, but that data will only be useful if it is interpreted properly. Each IoT device has a unique identity. This identification is helpful in tracking the equipment and at times for querying its status.

3. Scalability

The number of elements connected to the IoT zone is increasing day by day. Hence, an IoT setup should be capable of handling the massive expansion.

CHARACTERISTICS OF IoT

4. Dynamic and Self-Adapting (Complexity)

IoT devices should dynamically adapt themselves to changing contexts and scenarios. Assume a camera meant for surveillance. It should be adaptable to work in different conditions and different light situations (morning, afternoon, and night).

5. Architecture

[IoT Architecture] cannot be homogeneous in nature. It should be hybrid, supporting different manufacturers ‘ products to function in the IoT network.

6. Safety

There is a danger of the sensitive personal details of the users getting compromised when all his/her devices are connected to the internet. Hence, data security is the major challenge.

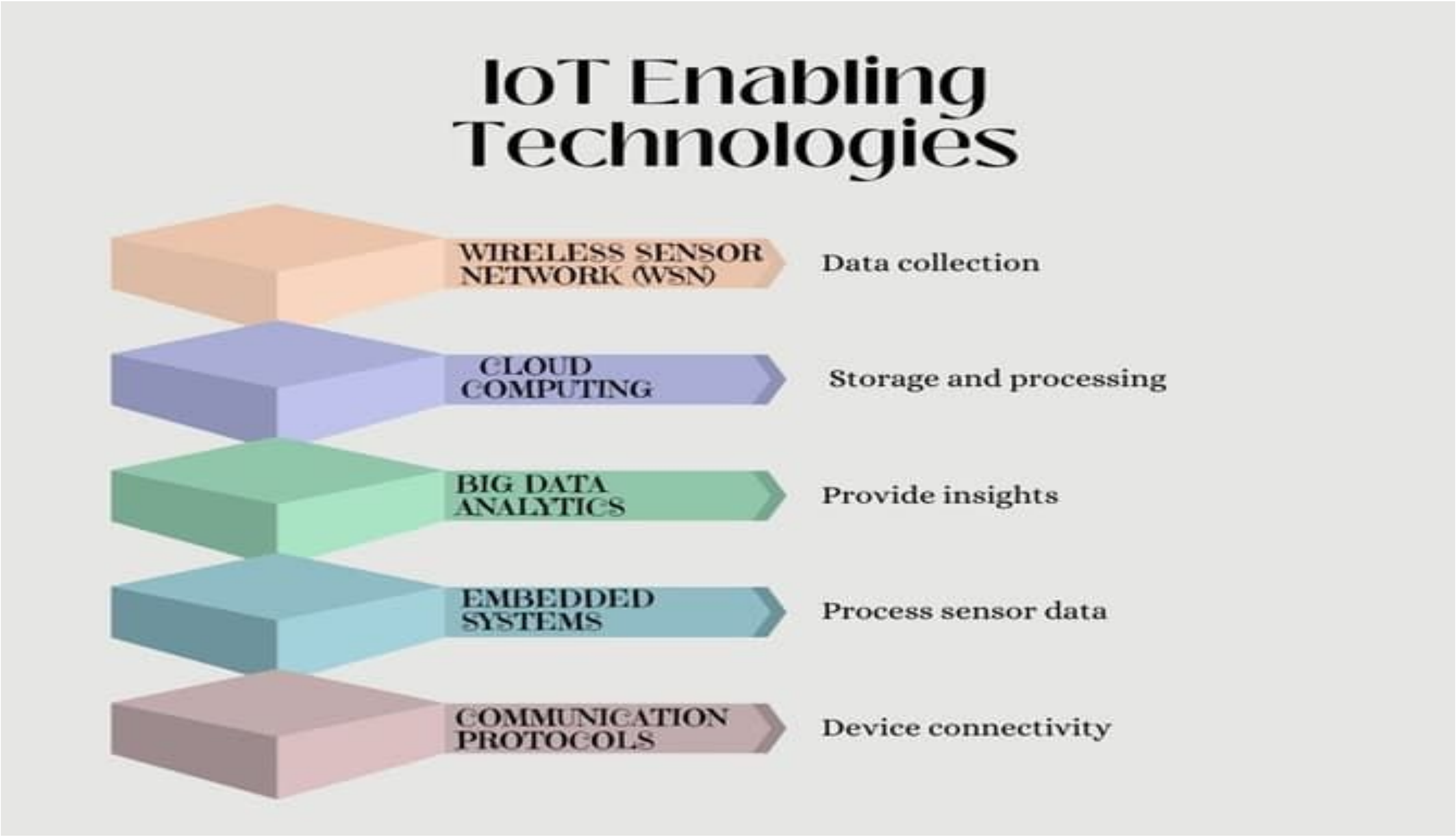
CHARACTERISTICS OF IoT

7. Self Configuring: This is one of the most important characteristics of IoT. IoT devices are able to upgrade their software in accordance with requirements with a minimum of user participation.

8. Interoperability: It refers to the ability of different IoT devices and systems to communicate and exchange data with each other, regardless of the underlying technology or manufacturer.To achieve interoperability, IoT devices, and systems use standardized communication protocols and data formats.

9. Embedded Sensors and Actuators: Embedded sensors and actuators are critical components of the Internet of Things (IoT). They allow IoT devices to interact with their environment and collect and transmit data.

10. Autonomous operation: Autonomous operation refers to the ability of IoT devices and systems to operate independently and make decisions without human intervention.


IoT Enabling Technologies
WIRELESS SENSOR NETWORK (WSN)
Data collection
CLOUD COMPUTING
Storage and processing
BIG DATA ANALYTICS
Provide insights
EMBEDDED SYSTEMS
Process sensor data
COMMUNICATION PROTOCOLS
Device connectivity

Internet of Things (IoT) Enabling Technologies

1. Wireless Sensor Network(WSN) :

A **WSN** comprises distributed devices with sensors which are used to monitor the environmental and physical conditions. A **wireless sensor network** consists of end nodes, routers and coordinators. End nodes have several sensors attached to them where the data is passed to a coordinator with the help of routers.

Example –

- Weather monitoring system
- Indoor air quality monitoring system
- Soil moisture monitoring system
- Surveillance system
- Health monitoring system

Internet of Things (IoT) Enabling Technologies

2. Cloud Computing :

It provides us the means by which we can access applications as utilities over the internet. Cloud means something which is present in remote locations.

With Cloud computing, users can access any resources from anywhere like databases, webservers, storage, any device, and any software over the internet.

IaaS (Infrastructure as a service**) :-**Ex : Web Hosting, Virtual Machine etc.
PaaS (Platform as a service**):-** Ex : App Cloud, Google app engine
SaaS (Software as a service**):-** Ex : Google Docs, Gmail, office etc.

The characteristics of cloud computing are:

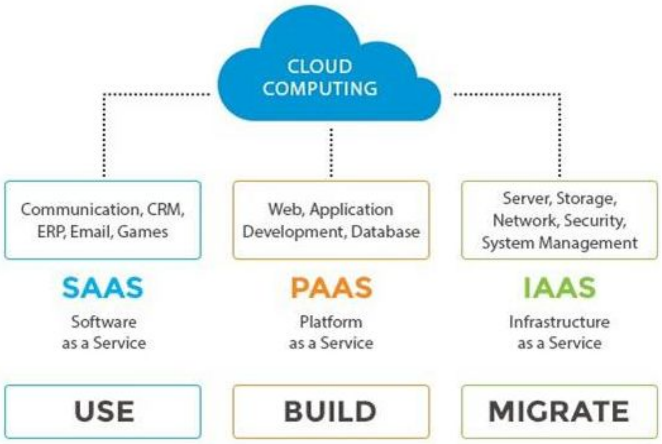
On demand: The resources in the cloud are available based on the traffic. If the incoming traffic increases, the cloud resources scale up accordingly and when the traffic decreases, the cloud resources scale down accordingly.

Autonomic: The resource provisioning in the cloud happens with very less to no human intervention. The resources scale up and scale down automatically.

Scalable: The cloud resources scale up and scale down based on the demand or traffic. This property of cloud is also known as elasticity.

Pay-per-use: On contrary to traditional billing, the cloud resources are billed on pay-per-use basis. You have to pay only for the resources and time for which you are using those resources.

Ubiquitous: You can access the cloud resources from anywhere in the world from any device. All that is needed is Internet. Using Internet you can access your files, databases and other resources in the cloud from anywhere.



Internet of Things (IoT) Enabling Technologies

3. Big Data Analytics :

It refers to the method of studying massive volumes of data or big data. Collection of data whose volume, velocity or variety is simply too massive and tough to store, control, process and examine the data using traditional databases.

Big data is gathered from a variety of sources including social network videos, digital images, sensors and sales transaction records.

Examples –

- Bank transactions
- Data generated by IoT systems for location and tracking of vehicles
- E-commerce and in Big-Basket
- Health and fitness data generated by IoT system such as a fitness bands

Internet of Things (IoT) Enabling Technologies

4. Communications Protocols :

They are the backbone of IoT systems and enable network connectivity and linking to applications. Communication protocols allow devices to exchange data over the network. Multiple protocols often describe different aspects of a single communication.

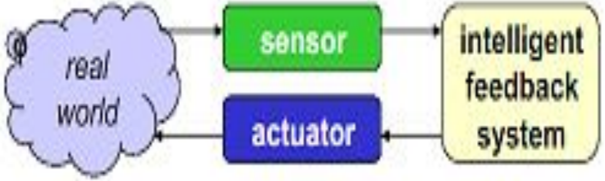
5. Embedded Systems :

It is a combination of hardware and software used to perform special tasks.
It includes microcontroller and microprocessor memory, networking units (Ethernet Wi-Fi adapters), input output units (display keyword etc.) and storage devices (flash memory).
It collects the data and sends it to the internet.

CONCEPT OF TRANSDUCERS/SENSORS

Transducers

- Transducer
  - a device that converts a primary form of energy into a corresponding signal with a different energy form
    - <u>Primary Energy Forms</u>: mechanical, thermal, electromagnetic, optical, chemical, etc.
  - take form of a **sensor** or an **actuator**
- Sensor (e.g., thermometer)
  - a device that detects/measures a signal or stimulus
  - acquires information from the "real world"
- Actuator (e.g., heater)
  - a device that generates a signal or stimulus

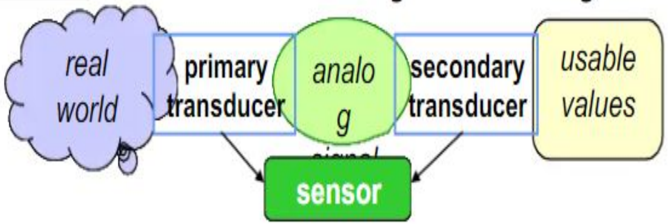


Sensor Systems

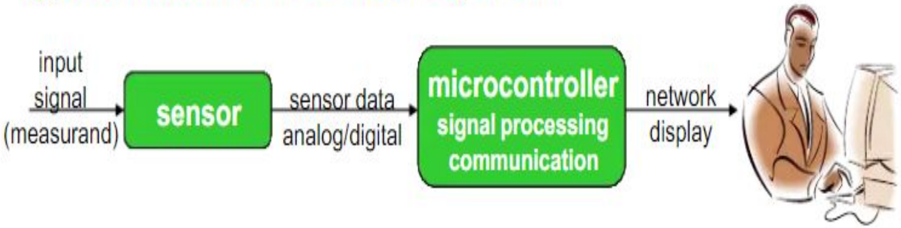
Typically interested in ***electronic sensor***

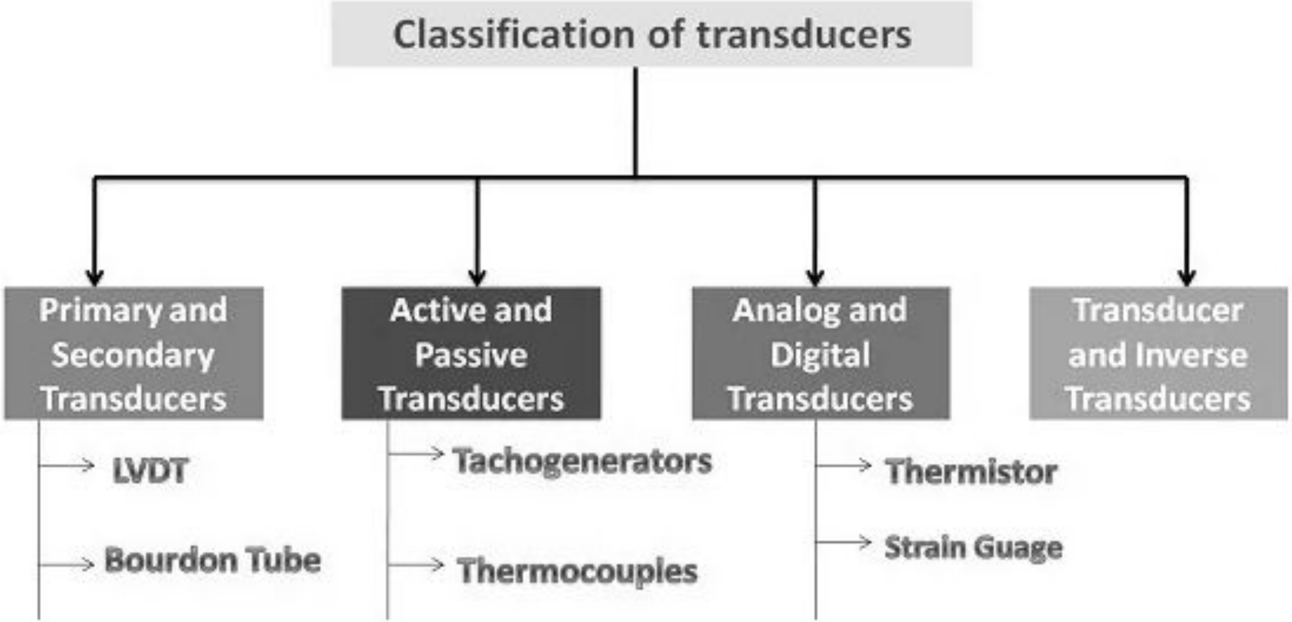
- convert desired parameter into electrically measurable signal

- General Electronic Sensor
  - <u>primary transducer</u>: changes "real world" parameter into electrical signal
  - <u>secondary transducer</u>: converts electrical signal into analog or digital values



- Typical Electronic Sensor System





Classification of transducers
Primary and Secondary Transducers
LVDT
Bourdon Tube
Active and Passive Transducers
Tachogenerators
Thermocouples
Analog and Digital Transducers
Thermistor
Strain Guage
Transducer and Inverse Transducers

Sensor Types:

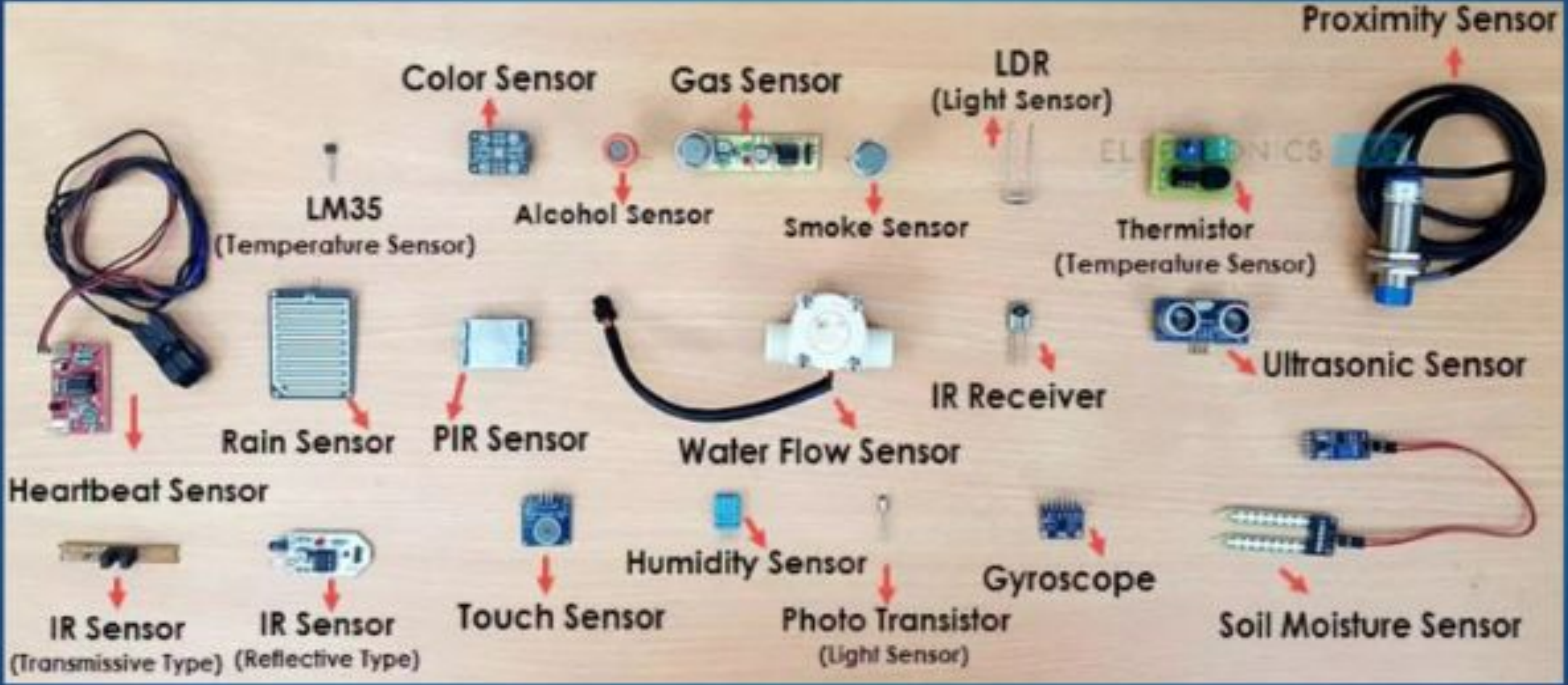
- Light Sensor
  1. IR Sensor (IR Transmitter / IR LED)
  2. Photodiode (IR Receiver)
  3. Light Dependent Resistor
- Temperature Sensor
  1. Thermistor
  2. Thermocouple
- Pressure/Force/Weight Sensor
  1. Strain Gauge (Pressure Sensor)
  2. Load Cells (Weight Sensor)
- Bio-Sensor
- Chemical Sensor

Sensor Types:

- Position Sensor
  1. Potentiometer
  2. Encoder
- Hall Sensor (Detect Magnetic Field)
- Flex Sensor
- Sound Sensor
- Microphone
- Ultrasonic Sensor
- Touch Sensor
- PIR Sensor
- Tilt Sensor
- Accelerometer
- Gas Sensor

Pictorial Depiction of Few Sensors:

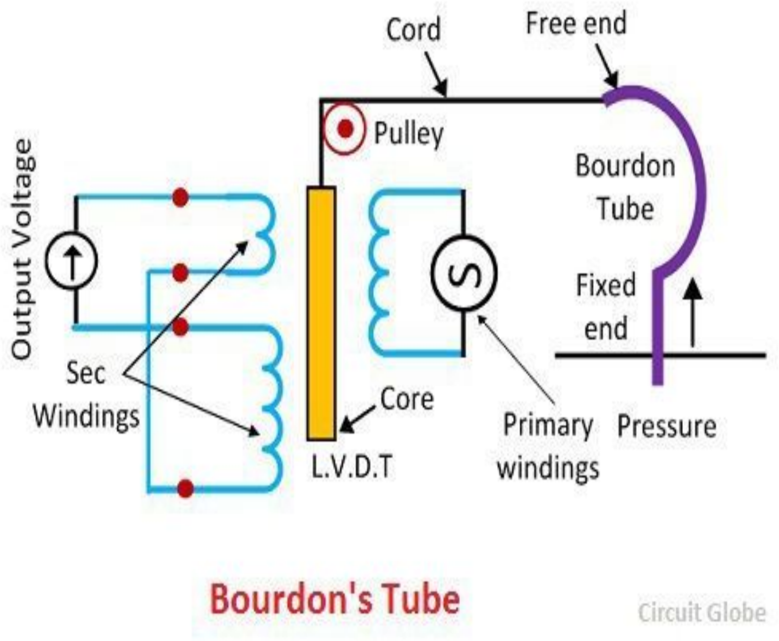
- Few Sensor Types



Primary and Secondary Transducers

On the basis of the role of the transducing element, transducers are divided into:

- **Primary transducers are generally the transducers that respond to external stimulation and general output for that signal. The output is generally a change in any factor affecting the secondary transducer.**
- **Secondary transducers are the ones that convert the output of the primary transducer into an electrical signal.**



The Bourdon's Tube is the primary transducer, and the L.V.D.T is called the secondary transducer.

DIFFERENCE BETWEEN ACTIVE AND PASSIVE TRANSDUCER

Active Transducer	Passive Transducer
The active transducer is also called as self generating type transducer.	The passive transducer is also called as externally powered transducer.
The active transducer does not require any auxiliary (external) power supply.	The passive transducer requires auxiliary (external) power supply for transduction.
The signal conversion is simpler.	The signal conversion is more complicated.
The energy required to produce output is obtained from the physical quantity.	They also derived part of the power required for conversion from physical quantity under measurement.
Example of active transducer is bourdon tube.	Example of passive transducer is LVDT (linear variable differential transformer).
It generates electric current or voltage directly in response to environmental stimulation.	It gives a change in some passive electrical quantity, such as capacitance, resistance or inductance, as a result of stimulation.

Active & Passive Sensors:

Active Sensors:

- ❖ The type of sensors that produces output signal without the help of external excitation supply.
- ❖ The own physical properties of the sensor vary with respect to the applied external effect.
- ❖ Therefore, it is also called as Self Generating Sensors.
- ❖ Any sensor which requires to input energy to the environment in order to retrieve the measurement is active.
- ❖ Examples: Photovoltaic cells, Thermocouples, Piezoelectric device.

Passive Sensors:

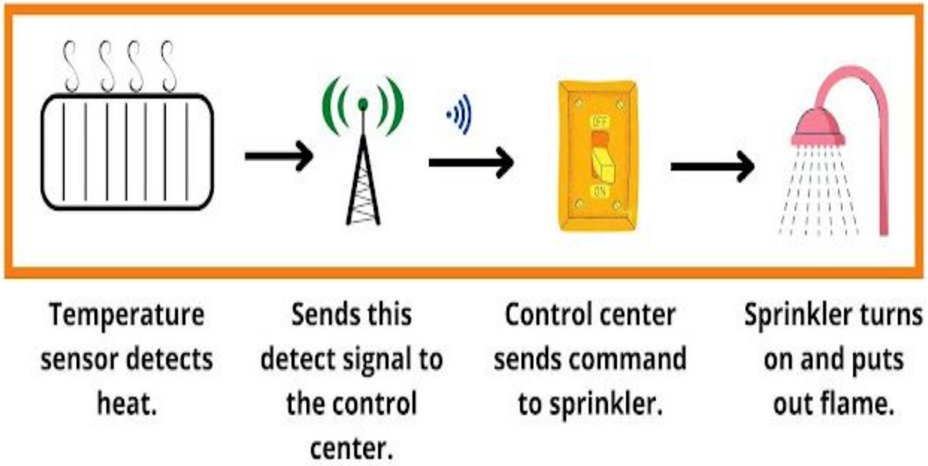
- ❖ The type of sensors that produces output signal with the help of external excitation supply.
- ❖ They need any extra stimulus or voltage.
- ❖ Sensors are able to retrieve a measurement without actively interacting with the environment.
- ❖ Example: Strain Gauge, Magnetometer, Barometer.

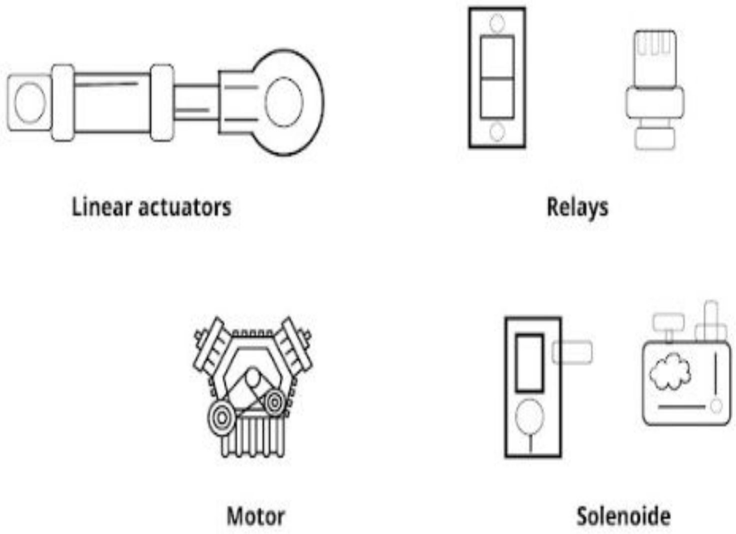
Analog & Digital Sensors:

- **Analog Sensors:** The sensor that produces continuous signal with respect to time with analog output is called as Analog sensors.
- The analog output generated is proportional to the measured or the input given to the system. Generally, analog voltage in the range of 0 to 5 V or current is produced as the output.
- The various physical parameters like temperature, stress, pressure, displacement, etc. are examples for continuous signals.
- **Digital Sensors:** When data is converted and transmitted digitally, it is called as Digital sensors.
- Digital sensors produce discrete output signals.
- Discrete signals will be non- continuous with time and it can be represented in "bits" for serial transmission and in "bytes" for parallel transmission.
- Digital output can be in form of Logic 1 or Logic 0 (ON or OFF). A digital sensor consists of sensor, cable and a transmitter.
- The measured signal is converted into a digital signal inside the sensor itself without any external component. Cable is used for long distance transmission.

What Exactly Are Actuators?

A device that turns electrical energy into mechanical energy is known as an actuator. An actuator, in other terms, is a component that can move or control a mechanism or system. Actuators are commonly employed in industrial automation, robotics, and other applications requiring precise mechanical system control.





Types of Actuators

Few Examples of Actuators in Internet of Things

Smart Home Systems: Actuators are an important part of smart home systems. They let customers remotely operate various equipment such as lights, heating systems, and security systems. A smart thermostat, for example, can use temperature sensors to change the temperature in a home, and an actuator can activate the heating or cooling system as needed.

Industrial Automation: Actuators are frequently employed in industrial automation to control machines and other systems. Hydraulic actuators, for example, might be used to control the movement of robotic arms in a factory, while electric actuators could be used to regulate the position of a conveyor belt.

Agriculture: Actuators are rapidly being employed to automate numerous processes in agriculture, such as irrigation and harvesting. An actuator, for example, can control the flow of water in an irrigation system or the position of a robotic arm used to harvest crops.

Healthcare: Actuators are employed in a variety of healthcare applications, including prostheses and medical equipment. An actuator, for example, can regulate the movement of a prosthetic limb or the location of a surgical tool during treatment.

Electrical Actuators

An electric actuator converts electricity into kinetic energy in either a linear (along a straight line), or rotary (in a circle) motion. The motor of an electric actuator can operate at any voltage, **Advantages:** cheap, clean, speedy type of actuator.• **Examples:** Solenoid based electric bell ringing mechanism

Thermal Actuators

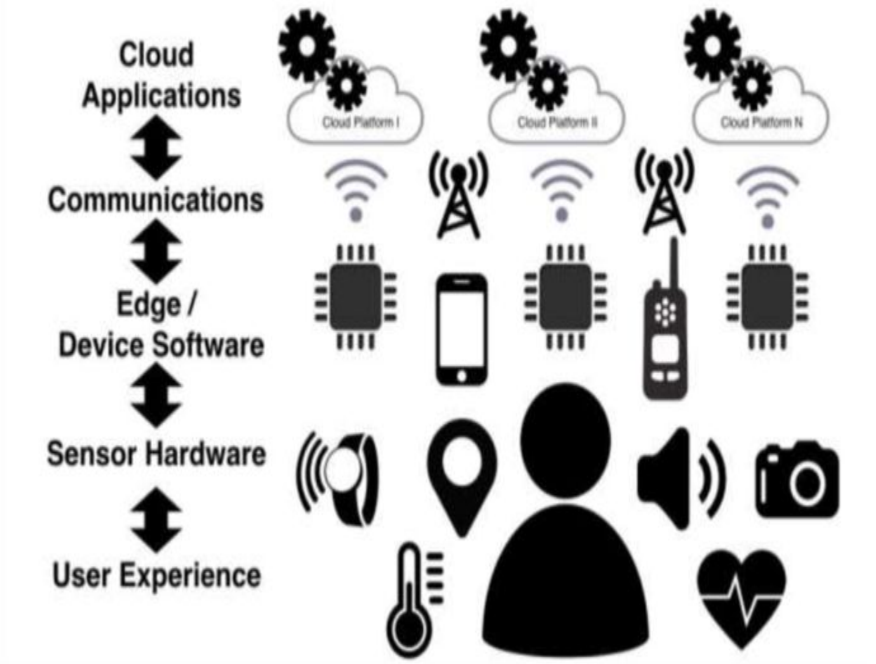
A thermal actuator is a type of non-electric motor. It’s equipped with thermal-sensitive material that’s capable of producing linear motion in response to temperature changes. Temperature changes can give rise to such tasks as releasing latches, operating switches, and opening or closing valves. **Examples:** Thermal actuator is thermostat,

Mechanical Actuators

Mechanical actuators convert one form of motion into another, they utilize gears, chains, pulleys, rails, and different contraptions for their operations. They are often combined with another actuator and driving mechanism. They are used for increasing torque or power of the output or even to convert linear motion to rotary or vice versa. **Examples:** Rack and pinion mechanism and Crank shaft

IoT Stack

- ❖ Even the most basic IoT devices need a range of technologies to function.
- ❖ Sensors, actuators, and computers use software to transmit data through a network to another device or application and vice versa.
- ❖ Collectively, this technology is known as the IoT stack—and it looks different for every IoT device.
- ❖ The IoT stack as the following four layers:
  - ➢ Hardware
  - ➢ Software
  - ➢ Protocols
  - ➢ Cloud



The first layer: hardware

- ❖ The hardware layer of your IoT stack encompasses the physical components of your device, such as sensors, mainboards, modules, actuators, SIM cards, antennae, and other physical pieces.
- ❖ This layer includes everything built into or attached to your device, as well as any other physical parts required for the operation of your device, such as modems, IoT routers, and IoT gateways.
- ❖ Your hardware governs the kinds of data your device can collect, what it can do with that data, and to a degree, what firmware and protocols it can utilize.
- ❖ In IoT, the hardware layer is often intended to last for the lifetime of the device, and the quality of these components can have a significant impact on your device's lifespan.

The second layer: device software

- The device software layer of your IoT stack includes components such as your device's operating system (OS), the application that interacts with your device and its data, and any firmware needed for your device's OS to interact with its hardware.
- Depending on the hardware and protocols in your IoT stack, some of the tech in this layer can be upgraded through over-the-air updates. This is vital for closing holes in security that emerge over time.

The third layer: protocol stack

- ❖ Protocols are essentially the languages that devices and network entities use to communicate.
- ❖ The protocol stack defines what kind of data your hardware can transmit and receive, how those transmissions will be secured (if at all), what kind of network the device will use to communicate, how network entities verify the accuracy of a transmission, and what your solution will prioritize (such as speed versus accuracy).
- ❖ Network entities need to use the same protocols to understand each other. Otherwise, their transmissions are like two humans trying to have a conversation when they don't speak the same language.
- ❖ An IoT device won't always have the data throughput to use the security protocols and other standards your solution needs, which may make it unable to communicate directly with your application. In these instances, your solution may use an IoT gateway as an intermediary to "translate" information from one entity into the protocols the other entity understands. This is common in use cases like smart meter communication.
- ❖ There may be a regulating body that keeps the protocol up-to-date and ensures it's standardized globally, or it may be open source with no enforced universal standard.
- ❖ Your IoT stack uses protocols at every layer of the network, and the hardware and software that comprise your solution may rely on different protocols for different interactions and functions. As your hardware and software changes and your device gains new functionality, your IoT stack may need to incorporate new protocols as well and/or discard ones that no longer meet your needs.

The fourth layer: the cloud

- It's a combination of scalable, globally accessible data storage, management, analytics, and utilization solutions.
- Whether you fully rely on cloud service providers like Amazon Web Services, Microsoft Azure, and Google Cloud Platform, or you build some portion of your cloud solution yourself, "the cloud" enables you to store all of the data from all of your IoT devices on cloud-based infrastructure, and leverage cloud-based applications to give your solution advanced capabilities like machine learning and artificial intelligence.
- "Big data" is one of the things that makes IoT solutions so valuable and effective—and accessibility is another.
- The cloud layer of your tech stack is what brings these benefits to your device.

IoT LEVELS

Device :

An IoT device allows identification, remote sensing, monitoring capabilities. Remote

Resource:

• Software components on the IoT device for -accessing, processing and storing sensor information, -controlling actuators connected to the device. - enabling network access for the device.

Controller Service:

• Controller service is a native service that runs on the device and interacts with the web services.

•It sends data from the device to the web service and receives commands from the application (via web services) for controlling the device.

Database:

•Database can be either local or in the cloud and stores the data generated by the IoT device.

Web Service:

•Web services serve as a link between the IoT device, application, database and analysis components.

•It can be implemented using HTTP and REST principles (REST service) or using the WebSocket protocol (WebSocket service).

Analysis Component:

• Analysis Component is responsible for analyzing the IoT data and generating results in a form that is easy for the user to understand.

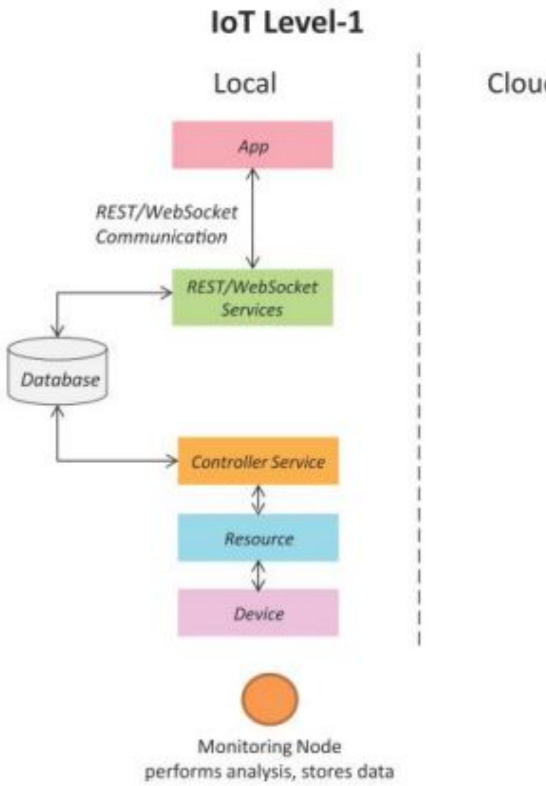
Application:

•IoT applications provide an interface that the users can use to control and monitor various aspects of the IoT system.

•Applications also allow users to view the system status and the processed data.

IoT Level-1

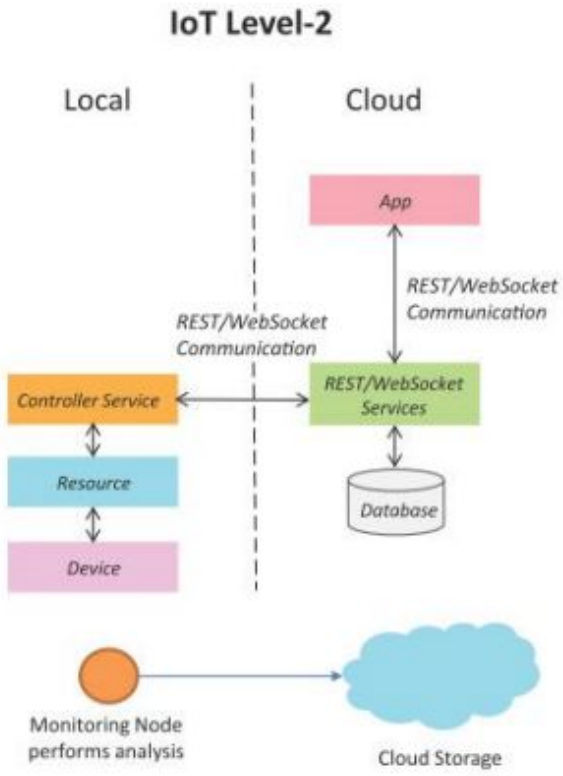
- A level-1 IoT system has a single node/device that performs sensing and/or actuation, stores data, performs analysis and hosts the application
- Level-1 IoT systems are suitable for modeling low-cost and low-complexity solutions where the data involved is not big and the analysis requirements are not computationally intensive.



IoT – Level 1 Example : Home Automation System

IoT Level-2

- A level-2 IoT system has a single node that performs sensing and/or actuation and local analysis.
- Data is stored in the cloud and application is usually cloud-based.
- Level-2 IoT systems are suitable for solutions where the data involved is big, however, the primary analysis requirement is not computationally intensive and can be done locally itself.

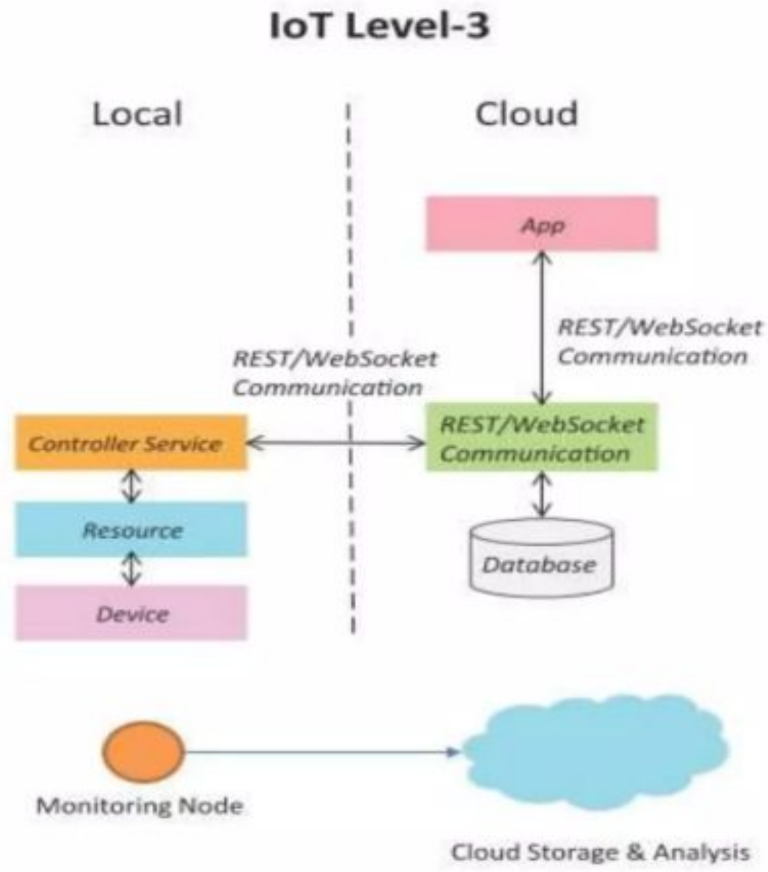


IoT – Level 2 Example: Smart Irrigation

IoT Level-3

A level-3 IoT system has a **single node**. Data is stored and **analyzed in the cloud** and the **application is cloud-based.**

Level-3 IoT systems are suitable for solutions where the **data involved is big** and the **analysis requirements are computationally intensive**.



IoT – Level 3 Example: Tracking Package Handling

Sensors used

Accelrometer

sense movement or vibrations

Gyroscope

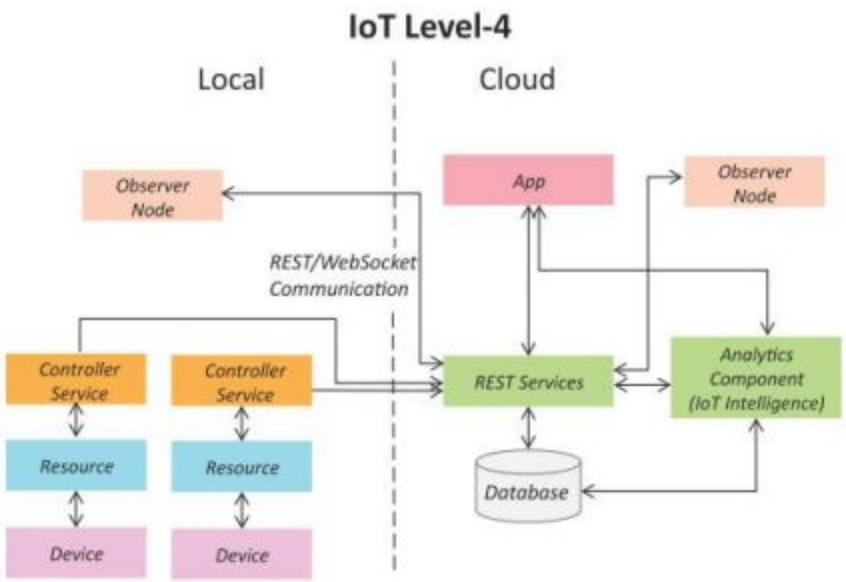
Gives orientation info

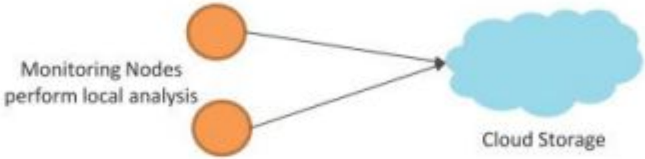
IoT Level-4

A level-4 IoT system has **multiple nodes** that perform **local analysis**. **Data is stored in the cloud** and the application is cloud-based.

Level-4 contains local and cloud-based **observer nodes** which can subscribe and receive information collected in the cloud from IoT devices.

Level-4 IoT systems are suitable for solutions where **multiple nodes are required**, the **data involved is big** and the **analysis requirements are computationally intensive.**



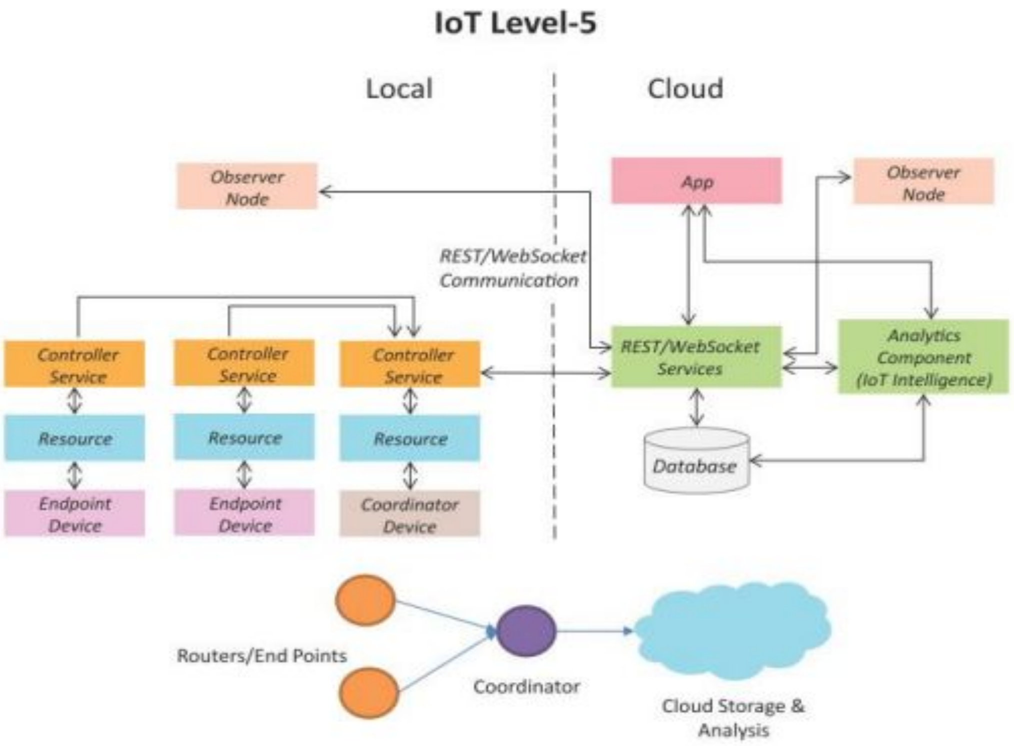


IoT – Level 4 Example: Noise Monitoring

Sound Sensors are used

IoT Level-5

- A level-5 IoT system has **multiple end nodes** and **one coordinator node.**
- The end nodes perform sensing and/or actuation.
- The coordinator node **collects data from the end nodes and sends it to the cloud.**
- Data is stored and **analyzed in the cloud** and the **application is cloud- based.**

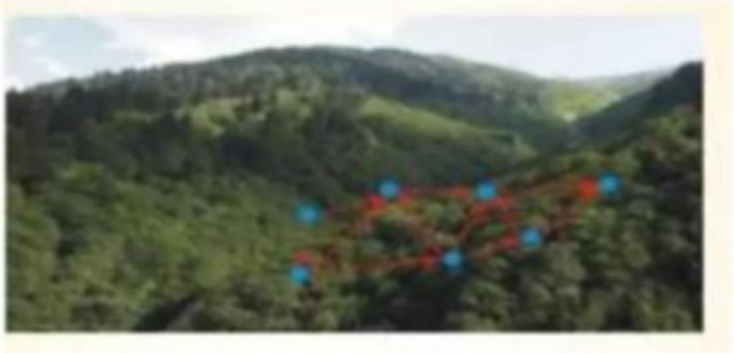


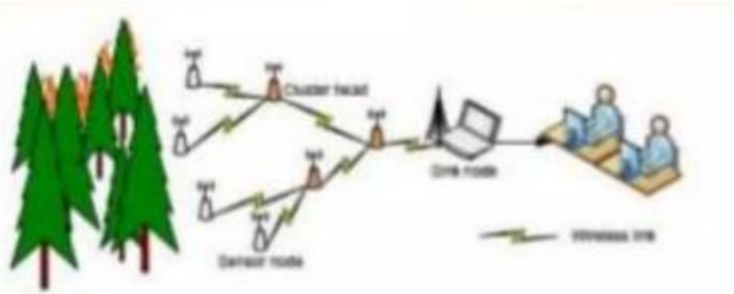
Level-5 IoT systems are suitable for **solutions based on wireless sensor networks**, in which the **data involved is big** and the **analysis requirements are computationally intensive.**

IoT – Level 5 Example: Forest Fire Detection

Detect forest fire in early stages to take action while the fire is still controllable.

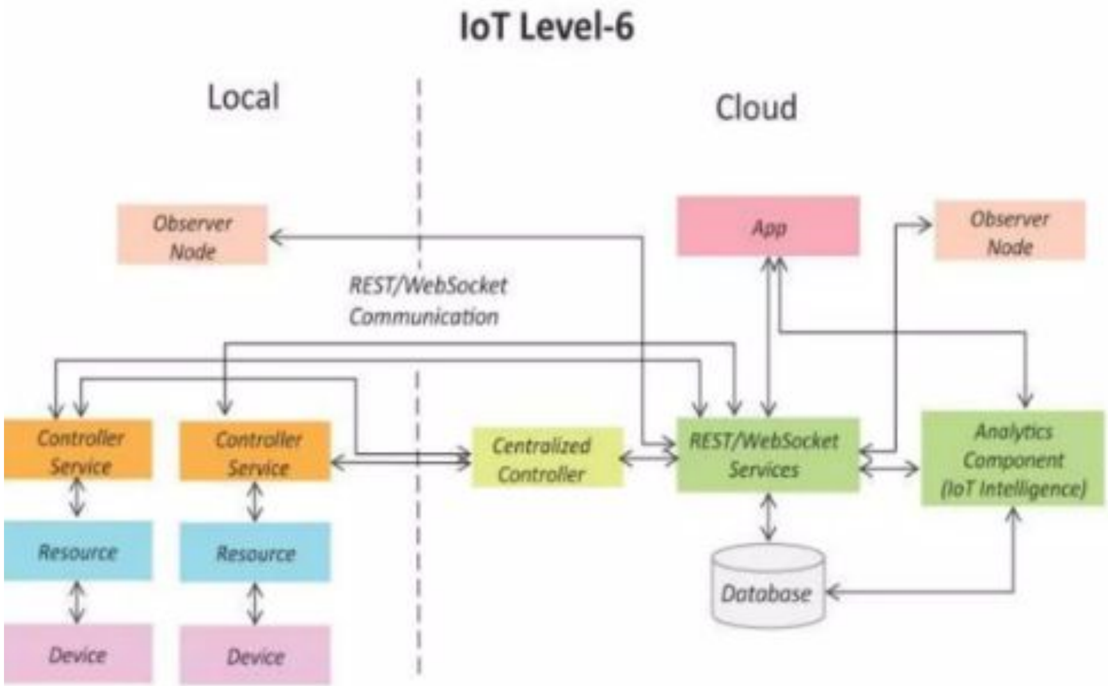
Sensors measure the temperature, smoke, weather, slope of the earth, wind speed, speed of fire spread, flame length

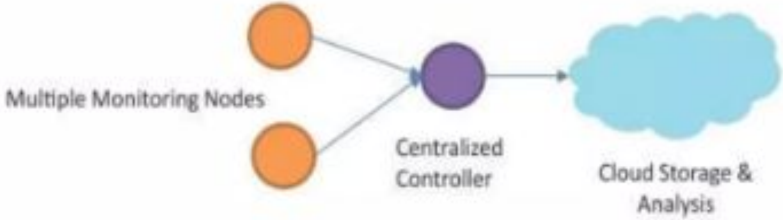




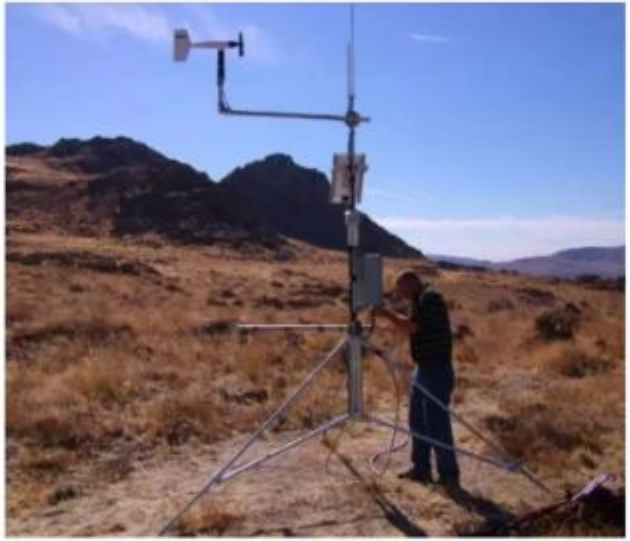
IoT Level-6

- A level-6 IoT system has **multiple independent end nodes** that perform sensing and/or actuation and send data to the cloud.
- **Data is stored in the cloud** and the **application is cloud-based**.
- The **analytics** component analyzes the data and stores the results in the **cloud database**.
- The results are visualized with the cloud-based application.
- The **centralized controller** is aware of the status of all the end nodes and **sends control commands to the nodes**.





IoT – Level 6 Example: Weather Monitoring System



Sensors used

- Wind speed and direction
- Solar radiation
- Temperature (air, water, soil)
- Relative humidity
- Precipitation
- Snow depth
- Barometric pressure
- Soil moisture